Ex: 14
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write a code using RAW Sockets to implement packet sniffing

Aim:

To implement a simple packet sniffer in JAVA that captures and displays network packets

Algorithm:

* start the program
* create a RAW Socket using (AF_PACKET, SOCK_RAW)
* Bind the Socket to the required network interface.
* continuously receive packet from the network.
* Extract the Ethernet header from each packet.
* IPV4 packets, extract the source and destination IP address
* Display the packet information.
* Repeat until the user terminates the program.
* close the socket.

Program:

```
import Socket
import struct
raw_socket = Socket.socket(Socket.AF_PACKET
    , Socket.SOCK_RAW, Socket.
    ntohs(3))
print("packet sniffer started....")
print("waiting for packets...\n")
```

```
try:
    while True:
        packet, address = raw_socket.recvfrom(65535)
        ethernet_header = packet[1:4]
        dest_mac, src_mac, protocol = struct.
                                unpack("! 6s6s4", ethr_header)
        dest_mac = ":".join("%02x" % byte for
                                byte in dest_mac)
        src_mac = ":".join("%02x" % byte for
                                byte in src_mac)
        print("=========")
        print("Source MAC:", src_mac)
        print("Destination MAC:", dest_mac)
        print("protocol:", hex(protocol))
        if protocol == 0x0800:
            ip_header = packet[14:34]
version_ihl, tos, total_length, identification,
flags_fragment, ttl, proto, checksum,
src_ip, dest_ip = struct.unpack
("!BBHHHBBH4s4s", ip_header)

SRC_IP = socket.inet_ntoa(src_ip)
dst_IP = socket.inet_ntoa(dest_IP)
```

sample outcome:

packet sniffer started

waiting for packets

Source MAC : 3C : 52 : 82 : 11 : 22 : 33

Destination MAC : ff : ff : ff : ff : ff : ff : ff

protocol : 0 x 306

packet Type : ARP

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source MAC : 3C : 52 : 82 : 11 : 22 : 33

Destination MAC : 84 : 16 : 64 : 44 : 55 : 66

protocol : 0 x 800

Source IP : 192 . 168 . 1 . 10

Destination IP : 192 . 168 . 1 . 1

TTL : 64

IP protocol : 6

```
elif protocol = 0x0806:
    print("packet Type: ARP")
elif protocol == 0x86DD:
    print("packet Type: IPV6")

else:
    print("packet Type: other")
except KeyboardInterrupt:
    print("\n packet Sniffing stopped.")

finally:
    raw_socket.close()
```

Result:-

Thus, a packet_Sniffing program using RAW Socket was Successfull implemented. The program captures network packets & display important information such as source MAC address, destination MAC address, source IP address, destination IP address and TTL.